SRB POS API Offline Integration Guide

Version: 1.0
Document Type: Technical Integration Specification
Prepared For: POS Software Vendors and Integrators
Integration Mode: Offline Integration

---

1. Introduction

1.1 Purpose

This document defines the technical specifications required for integration between Point of Sale (POS) software and the SRB POS Connector through the Offline Integration API.

The API enables POS applications to submit invoice information to the SRB POS Connector, receive an SRB Invoice Number, and generate a QR Code for invoice verification.

---

2. System Overview

The integration consists of:

1. POS Software (Taxpayer's System)
2. SRB POS Connector (SRB's software installs on Taxpayer's System)
3. SRB Verification System (SRB End)

Process Flow

1. Cashier generates invoice in POS.
2. POS sends invoice payload to SRB POS Connector.
3. Connector validates invoice data.
4. Connector generates SRB Invoice Number.
5. Connector returns QR Code URL.
6. POS prints invoice with QR Code.

---

3. API Endpoint

URL

http://localhost:8282/pos/SalesInvoiceServices

Method

POST

Content Type

application/json

---

4. Sample Invoice Request


```
{
  "posId":331,
  "name":"HEAD OFFICE",
  "ntn":"8000001",
  "invoiceDateTime":"2025-12-26 23:58:58",
  "invoiceType":1,
  "invoiceId":"PZ010",
  "rateValue":15,
  "saleValue":5000,
  "taxAmount":750,
  "discountAmount":0,
  "serviceCharges":0,
  "extraCharges":0,
  "netAmount":5750,
  "consumerName":"N/A",
  "consumerMobile":"N/A",
  "consumerEmail":"N/A",
  "consumerNTN":"N/A",
  "address":"N/A",
  "cpcCode":"N/A",
  "extraInf":"N/A",
  "modeOfPay":"Cash",
  "transType":"Test"
}
```


---

5. Successful Response


```
{
"srbInvoiceId":"331B2606KH10",
"resCode":"00",
"QRCodeLink":"https://apps.srb.gos.pk/InvoiceVerification/MobileInvoiceStatus
.jsp?invoiceVerification=331B2606KH10"
}
```


---

6. Error Responses

6.1 Missing Required Parameters


```
{
  "status":"BAD_REQUEST",
  "resCode":"01",
  "error":"Error in Validation",
  "Required Parameter(s) Missing":"netAmount, invoiceType",
  "Optional Parameter(s) Missing":"consumerEmail, cpcCode"
}
```

6.2 Invalid POS ID


```
{
  "status":"BAD_REQUEST",
  "resCode":"02",
  "error":"POS ID is not register"
}
```

6.3 Invalid Business Name


```
{
  "status":"BAD_REQUEST",
  "resCode":"02",
  "error":"Invalid Business/Restaurant Name"
}
```

6.4 Invalid NTN


```
{
  "status":"BAD_REQUEST",
  "resCode":"02",
```

```
  "error":"NTN provided is invalid"
}
```

6.5 Invalid Date Format

```
{
  "status":"BAD_REQUEST",
  "resCode":"02",
  "error":"Invalid Date Time Format is used for invoiceDateTime. It must be
in yyyy-MM-dd HH:mm:ss format"
}
```

6.6 Invalid Invoice Type

```
{
  "status":"BAD_REQUEST",
  "resCode":"02",
  "error":"Invalid Invoice Type. Use 1 for Normal Invoice or 2 for Sales
Return"
}
```

6.7 Tax Calculation Error

```
{
  "status":"BAD_REQUEST",
  "resCode":"02",
  "error":"Tax amount is not calculated as per rate and sales value."
}
```

6.8 Invalid Net Amount

```
{
  "status":"BAD_REQUEST",
  "resCode":"02",
  "error":"Invalid Net Amount is used."
}
```

6.9 Duplicate Invoice

```
{
  "status":"Fail",
  "resCode":"02",
```

```
  "error":"Sales invoice cannot be entered due to duplicate InvoiceId."
}
```


---

7. Field Definitions

Field Name	Data Type	Required	Description
posId	Integer	Yes	SRB Registered POS ID
name	String	Yes	Business Name (As per registration in SRB and without special characters)
ntn	String	Yes	NTN Number (Without leading S and without digit after Hyphen)
invoiceDateTime	String	Yes	Invoice Date Time (Must be yyyy-MM-dd HH:mm:ss format)
invoiceType	Integer	Yes	1=Invoice, 2=Sales Return
invoiceId	String	Yes	Unique Invoice Number
rateValue	Number	Yes	Tax Rate
saleValue	Number	Yes	Gross Sale Amount
taxAmount	Number	Yes	Calculated Tax
discountAmount	Number	Yes	Discount Amount (If applicable otherwise 0)
serviceCharges	Number	Yes	Service Charges (If applicable otherwise 0)
extraCharges	Number	Yes	Extra Charges (If applicable otherwise 0)
netAmount	Number	Yes	Final Payable Amount
consumerName	String	No	Customer Name
consumerMobile	String	No	Customer Mobile
consumerEmail	String	No	Customer Email
consumerNTN	String	No	Customer NTN/CNIC
address	String	No	Customer Address
cpcCode	String	No	CPC Code
extraInf	String	No	Additional Information
modeOfPay	String	Yes	Cash/Card
transType	String	Yes	Test/Live

---

8. Calculation Rules

Tax Calculation

Formula:

taxAmount = (saleValue+serviceCharges+extraCharges) x (rateValue/100)

Example:

saleValue = 5000

serviceCharges = 10 (If applicable otherwise 0)

extraCharges = 10 (If applicable otherwise 0)

rateValue = 15

taxAmount = (5000 + 10 + 10) x (15/100)

taxAmount = 753

Net Amount Calculation

Formula:

netAmount = (saleValue + serviceCharges + extraCharges + taxAmount) - discountAmount

Example:

saleValue = 5000

taxAmount = 753

discountAmount = 50 (If applicable otherwise 0)

serviceCharges = 10 (If applicable otherwise 0)

extraCharges = 10 (If applicable otherwise 0)

netAmount = 5723

---

9. Validation Rules

The following validations are performed by the API:

1. POS ID must be registered.

2. NTN must be valid (Must be without leading S and without digit after Hyphen)
3. Invoice Id must be unique.
4. Invoice Date Time format must be:

yyyy-MM-dd HH:mm:ss

5. Sales value cannot be zero.
6. Sales value cannot be negative.
7. Tax amount must match the tax formula.
8. Net amount must be calculated correctly.
9. Invoice Type must be:
    - 1 = Normal Invoice
    - 2 = Sales Return
10. Required fields cannot be null or empty.

---

10. Optional Field Guidelines

Optional String Fields:

- consumerName
- consumerEmail
- consumerNTN
- consumerMobile
- address
- cpcCode
- extraInf

Can be sent as:

"N/A"

or

""

Optional Numeric Fields should be sent as:

0

---

11. QR Code Requirements

The QRCodeLink returned by the API must be converted into a QR Code and printed on the invoice.

Example:

SRB Invoice No: 331B2606KH10

The QR Code should encode the URL returned in the QRCodeLink field.

---

12. Sample Invoice Layout

Business Name: HEAD OFFICE

Invoice No: PZ010

SRB Invoice No: 331B2606KH10

Date: 2025-12-26 23:58:58

Sale Value: 5000.00

Tax Amount: 750.00

Net Amount: 5750.00

[QR CODE]

Thank You

---

13. Troubleshooting

Problem	Solution
POS ID Invalid	Verify POS registration (Use POS ID generated from SRB POS Portal)
NTN Invalid	Verify NTN configuration (Check whether NTN is without S and without digit after Hyphen)
Duplicate Invoice	Generate unique invoice number
Date Format Error	Use yyyy-MM-dd HH:mm:ss
Tax Calculation Error	Recalculate tax
Net Amount Error	Verify formula

Problem	Solution
Missing Parameters	Send all required fields

15. Best Practices

1. Generate unique invoice IDs.
2. Store SRB Invoice IDs permanently.
3. Print QR codes on all invoices.
4. Validate calculations before API submission.
5. Test thoroughly using Test mode.
6. Log all requests and responses.
7. Handle API errors gracefully.

16. QR Code Embedding

The generated invoice must include SRB Invoice Id and its respective link received from API must be embedded in QR code. The sample format for compliance is given below:

SRB Invoice No:

331B2606KH10